

Analysis of National Cranberry Cooperative, 1996

Lerner MBA · BUAD631 · Group Project · Spring 2026

Group Members

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16,746

bbls received
Sept 25, 1995

186%

dryer utilization
at 80% wet

42

peak trucks waiting under current
system

Our Recommendations

Dryers

Purchase 2 Additional Dryers (\$120,000)

- Current 3 dryers: utilization = 186% → severe bottleneck
- Adding 2 dryers cuts peak waiting trucks: 42 → 0
- Reduces dryer-area processing time from 22.33 hrs to 13.40 hrs.
- 1st dryer NPV = **\$3,767,435**;
2nd dryer NPV = **\$1,186,299**
Combined NPV = **\$4,953,735** at 10% discount rate.
- Both dryers have positive incremental NPV and eliminate truck waiting.

Holding Bins

Convert 0 bins initially

- With 5 dryers, max inventory = 2,313 bbls
- Wet bin capacity = 3,200 bbls → no overflow
- 0 bin conversions needed with 5 dryers
- Existing wet-bin capacity is sufficient after dryer expansion.

Labor Schedule

Two Full Shifts 7am–3pm and 3pm–11pm

- 53 total employees allocated across the 7-day peak week with each employee only scheduled for 5 days/week; ~19 workers per shift.
- Base cost: 19 workers × 2 shifts × 8h × \$15 = \$4,560/day
- With 5 dryers: zero hours OT beyond 11 pm
- Current system: Production can only run a maximum of 22 hours a day, so OT is capped at 6h beyond 11pm → 6h × 19 workers × \$22.50 = \$2,565 OT/day
- Two-shift plan reduces overtime risk on peak days.

Simulation Results: Bottleneck and Capacity Analysis

⚠ Bottleneck Identified			
Process	Demand/hr	Capacity/hr	Util.
Kiwanee Dumpers	1,396	3,000	47%
Wet Dechaffing	1,117	3,000	37%
Dry Destoning	279	4,500	6%
DRYERS (×3) bottleneck	1,117	600	186% ←
Separators(×3) (watch point)	1,396	1,200	116%
Avg wet: 1117 bbls/hr vs dryer cap: 600 bbls/hr = 186%			

Note - Separator capacity becomes the next downstream watch point after dryer expansion, but the required simulation analysis focuses on the dryer area.

📋 Labor Cost Detail
Base two-shift cost: 19 workers × 2 shifts × 8h × \$15 = \$4,560/day
OT cost rate: 19 workers × \$22.50/hr = \$427.50/hr
Sim 1: \$4,560 + 6h OT = \$7,125
Sim 2: \$4,560 + 1h OT = \$4,987.50
Sim 3: \$4,560 + zero OT = \$4,560

Full shifts are paid for 8 hours even if processing finishes earlier; with 5 dryers, wet berries clear before 11pm, but the second shift is still paid as a full shift.

OT hours are rounded/capped based on 11pm–5am operating limit.

Metric	Sim 1 3 Dryers (Current)	Sim 2 4 Dryers (+1)	Sim 3 5 Dryers (+2)
Dryer Capacity (bbls/hr)	600	800	1,000
Max Inventory Buildup (bbls)	6,316	4,243	2,313
Wet Bin Capacity (bbls)	3,200	3,200	3,200
Peak Trucks Waiting	42	14	0
Total Truck-Hours Waited	236	60	0
Processing time needed (hrs)	22.33	16.75	13.40
Bin Conversions Needed	13	5	0
Daily Labor Cost	\$7,125	\$4,987.50	\$4,560
NPV of Adding This Dryer	—	\$3,767,435	\$1,186,299

Red = current • Gold = +1 dryer • Green = +2 dryers
See Exhibits 2–4 for full hourly simulation detail

Financial Analysis: NPV Decision

Key Assumptions

Dryer cost: **\$60,000**

Bin conversion: **\$10,000/bin**

Depreciation: **12.5%/yr, 8 yrs**

Salvage value: **\$0**

Tax rate: **20%**

Discount rate: **10%**

Workforce: **Workforce: 53 total employees allocated across peak week; ~19 workers/shift**

Wage: **\$15/hr**

Overtime: **1.5× after 8h/day**

Truck wait cost: **\$100/hr (Case fn.1)**

Peak equivalent days/yr: **36.35 days/yr**

Wet berries: **80% (next year)**

Peak equivalent days/yr: **36.35 days/yr**
[(610,185 – 18,257 + 16,746) × 80%] ÷ 13,397 = 36.35 days/yr

Note - Peak equivalent days/year = annual wet berry volume ÷ wet berries processed on the representative peak day

NPV 1: 1st Additional Dryer

Simulation 2 vs Simulation 1

NPV

\$3,767,435

✓ **POSITIVE → RECOMMEND**

Investment (Year 0): **~~\$-60,000~~**

Annual labor savings: **\$77,698**

Annual truck savings: **\$639,760**

Annual depr. tax savings: **\$1,500**

Annual maint. cost: **~~\$600-\$2,400/yr~~**

→ Full cash flow table: Exhibit 5

NPV 2: 2nd Additional Dryer

Simulation 3 vs Simulation 2

NPV

\$1,186,299

✓ **POSITIVE → RECOMMEND**

Investment (Year 0): **~~\$-60,000~~**

Annual labor savings: **\$15,540**

Annual truck savings: **\$218,100**

Annual depr. tax savings: **\$1,500**

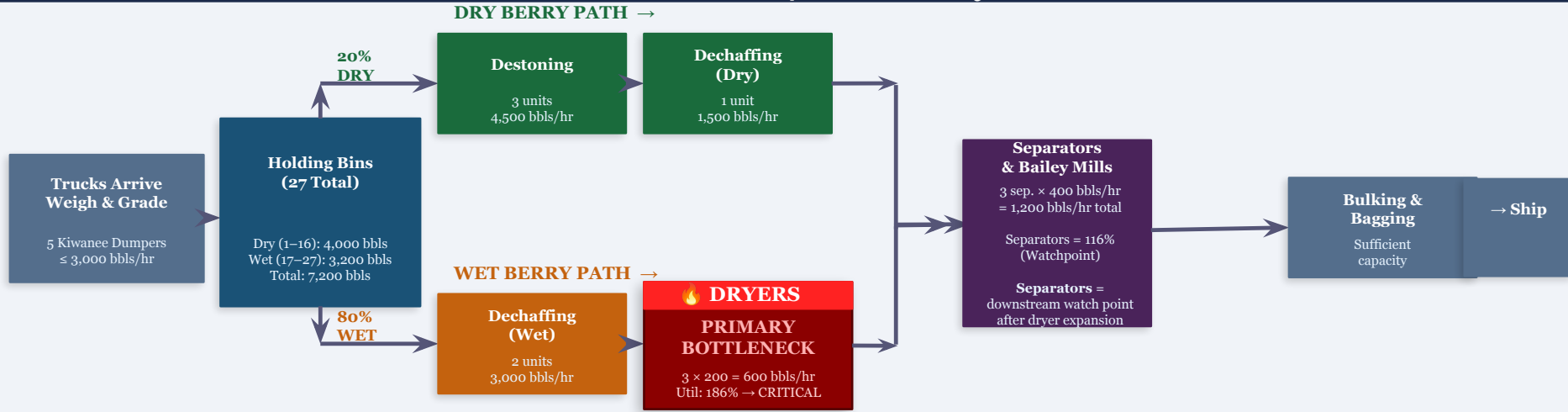
Annual maint. cost: **~~\$600-\$2,400/yr~~**

→ Full cash flow table: Exhibit 6

Labor model: peak staffing averages ~19 workers per shift; OT after 11pm applies only to the second-shift crew.

Bin conversion cost is not included in the final two-dryer NPV because the recommended option requires 0 bin conversions. Conversion costs would apply only to alternatives that actually convert bins.

Exhibit 1 — Process Flow Diagram: Receiving Plant #1 (80% Wet / 20% Dry)



Capacity & Bottleneck Analysis (80% wet: avg wet demand = 1,117 bbls/hr; avg total demand = 1,396 bbls/hr)

Process Step	# Units	Unit Cap. (bbls/hr)	Total Cap. (bbls/hr)	Demand (bbls/hr)	Implied Util.	Status
Kiwanee Dumpers	5	600	3,000	1,396	47%	✓ OK
Wet Dechaffing	2	1,500	3,000	1,117	37%	✓ OK
Dry Destoning	3	1,500	4,500	279	6%	✓ OK
🔥 DRYERS	3	200	600	1,117	186%	⚠️ BOTTLENECK
Separators	3	400	1,200	1,396	116%	⚠️ Watch point

Exhibit 2 — Simulation 1: Current — 3 Dryers

Dryer Capacity: 600 bbls/hr

Time Period	Total Arrivals (bbls)	Wet Arrivals (80% of total)	Wet bbls Processed	Inventory (undried bbls)	Bins Full? (> 3,200 bbls)	Trucks Waiting
7–8am	1,820	1,456	600	856	No	—
8–9am	1,749	1,399	600	1,655	No	—
9–10am	1,781	1,425	600	2,480	No	—
10–11am	1,716	1,373	600	3,253	YES ⚠	1
11am–12	1,514	1,211	600	3,864	YES ⚠	9
12–1pm	1,154	923	600	4,187	YES ⚠	14
1–2pm	1,530	1,224	600	4,811	YES ⚠	22
2–3pm	1,627	1,302	600	5,513	YES ⚠	31
3–4pm	1,181	945	600	5,858	YES ⚠	36
4–5pm	1,231	985	600	6,243	YES ⚠	41
5–6pm	841	673	600	6,316	YES ⚠	42
6–7pm	602	482	600	6,198	YES ⚠	40
TOTAL BY 7PM / MAX	16,746	13,398	7,200	Max: 6,316	Max: 42	

Summary Statistics	
Dryer capacity:	600 bbls/hr
Wet bin capacity:	3,200 bbls
Max inventory:	6,316 bbls
Bin overflow:	3,116 bbls
Peak trucks waiting:	42
Total truck-hrs waited:	236
Processing time required:	22.33 (Labor OT capped at 6hrs due to 5:00am hard stop)
Total Labor cost:	\$7,125
= 19 workers × 2 shifts × 8h × \$15	
= \$4,560	
OT beyond 11 pm: 6.00h × 19 × \$22.50:	
= \$2,565	
Bin conversions needed (to eliminate waiting):	13 bins

Exhibit 3 — Simulation 2: +1 Dryer (4 Total)

Dryer Capacity: 800 bbls/hr

Time Period	Total Arrivals (bbls)	Wet Arrivals (80% of total)	Wet bbls Processed	Inventory (undried bbls)	Bins Full? (> 3,200 bbls)	Trucks Waiting
7–8am	1,820	1,456	800	656	No	—
8–9am	1,749	1,399	800	1,255	No	—
9–10am	1,781	1,425	800	1,880	No	—
10–11am	1,716	1,373	800	2,453	No	—
11am–12	1,514	1,211	800	2,864	No	—
12–1pm	1,154	923	800	2,987	No	—
1–2pm	1,530	1,224	800	3,411	YES ⚠	3
2–3pm	1,627	1,302	800	3,913	YES ⚠	10
3–4pm	1,181	945	800	4,058	YES ⚠	12
4–5pm	1,231	985	800	4,243	YES ⚠	14
5–6pm	841	673	800	4,116	YES ⚠	13
6–7pm	602	482	800	3,798	YES ⚠	8
TOTAL BY 7 PM / MAX	16,746	13,398	9,600	Max: 4,243	Max: 14	

Summary Statistics	
Dryer capacity:	800 bbls/hr
Wet bin capacity:	3,200 bbls
Max inventory:	4,243 bbls
Bin overflow:	1,043 bbls
Peak trucks waiting:	14
Total truck-hrs waited:	60
Processing time required:	16.75 hrs
Total Labor cost:	\$4,987.50
= 19 workers × 2 shifts × 8h × \$15:	= \$4,560
OT beyond 11 pm: 1 h × 19 × \$22.50:	= \$427.50
Bin conversions needed (to eliminate waiting):	5 bins

Note - OT beyond 11pm = rounded to 1 hour because total processing time is 16.75 hours.

Exhibit 4 — Simulation 3: +2 Dryers (5 Total)

Dryer Capacity: 1000 bbls/hr

Time Period	Total Arrivals (bbls)	Wet Arrivals (80% of total)	Wet bbls Processed	Inventory (undried bbls)	Bins Full? (> 3,200 bbls)	Trucks Waiting	Summary Statistics	
7–8am	1,820	1,456	1,000	456	No	—	Dryer capacity:	1000 bbls/hr
8–9am	1,749	1,399	1,000	855	No	—	Wet bin capacity:	3,200 bbls
9–10am	1,781	1,425	1,000	1,280	No	—	Max inventory:	2,313 bbls
10–11am	1,716	1,373	1,000	1,653	No	—	Bin overflow:	None ✓
11am–12	1,514	1,211	1,000	1,864	No	—	Peak trucks waiting:	0
12–1pm	1,154	923	1,000	1,787	No	—	Total truck-hrs waited:	0
1–2pm	1,530	1,224	1,000	2,011	No	—	Processing time required:	13.40 hrs
2–3pm	1,627	1,302	1,000	2,313	No	—	Total Labor cost:	\$4,560
3–4pm	1,181	945	1,000	2,258	No	—	= 19 workers × 2 shifts × 8h × \$15:	= \$4,560
4–5pm	1,231	985	1,000	2,243	No	—	OT beyond 11 pm: zero h × 19 × \$22.50:	= \$0
5–6pm	841	673	1,000	1,916	No	—	Bin conversions needed (to eliminate waiting):	0 bins
6–7pm	602	482	1,000	1,398	No	—		
TOTAL BY 7 PM/ MAX	16,746	13,398	12,000	Max: 2,313	Max: 0			

Exhibit 5 — NPV Analysis: 1st Additional Dryer (Sim 2 vs Sim 1)

Assumptions	
Comparison:	Sim 2 vs Sim 1
Investment (Yr 0):	\$-60,000
Depreciation:	12.5%/yr × 8 yrs
Tax rate:	20%
Discount rate:	10%
Ann. labor savings:	\$77,698
Ann. truck savings:	\$639,760
Maintenance (Yr 1–4):	1–3% of invest.
Maintenance (Yr 5–8):	3–4% of invest.
Salvage value:	\$0

Year	Invest. (Yr 0)	Depr. Tax Sav.	Maint. Cost	Labor Savings	Truck Savings	Net Cash Flow	PV Factor	PV of Cash Flow
0 (now)	\$-60,000	—	—	—	—	\$-60,000	1.000	\$-60,000
1	—	\$1,500	\$600	\$77,698	\$639,760	\$718,358	0.909	\$653,053
2	—	\$1,500	\$1,200	\$77,698	\$639,760	\$717,758	0.826	\$593,192
3	—	\$1,500	\$1,200	\$77,698	\$639,760	\$717,758	0.751	\$539,259
4	—	\$1,500	\$1,800	\$77,698	\$639,760	\$717,158	0.683	\$489,827
5	—	\$1,500	\$1,800	\$77,698	\$639,760	\$717,158	0.621	\$445,298
6	—	\$1,500	\$1,800	\$77,698	\$639,760	\$717,158	0.564	\$404,815
7	—	\$1,500	\$2,400	\$77,698	\$639,760	\$716,558	0.513	\$367,709
8	—	\$1,500	\$2,400	\$77,698	\$639,760	\$716,558	0.467	\$334,282
NPV								\$3,767,435

NPV = Year 0 investment + present value of Year 1–8 cash flows discounted at 10%.

NPV = \$3,767,435 > \$0 → Future savings exceed the \$60K investment cost. RECOMMEND purchasing the first additional dryer.

Exhibit 6 — NPV Analysis: 2nd Additional Dryer (Sim 3 vs Sim 2)

Assumptions	
Comparison:	Sim 3 vs Sim 2
Investment (Yr 0):	\$-60,000
Depreciation:	12.5%/yr × 8 yrs
Tax rate:	20%
Discount rate:	10%
Ann. labor savings:	\$15,540
Ann. truck savings:	\$218,100
Maintenance (Yr 1–4):	1–3% of invest.
Maintenance (Yr 5–8):	3–4% of invest.
Salvage value:	\$0

Year	Invest. (Yr 0)	Depr. Tax Sav.	Maint. Cost	Labor Savings	Truck Savings	Net Cash Flow	PV Factor	PV of Cash Flow
0 (now)	\$-60,000	—	—	—	—	\$-60,000	1.000	\$-60,000
1	—	\$1,500	\$600	\$15,540	\$218,100	\$234,540	0.909	\$213,218
2	—	\$1,500	\$1,200	\$15,540	\$218,100	\$233,940	0.826	\$193,340
3	—	\$1,500	\$1,200	\$15,540	\$218,100	\$233,940	0.751	\$175,761
4	—	\$1,500	\$1,800	\$15,540	\$218,100	\$233,340	0.683	\$159,374
5	—	\$1,500	\$1,800	\$15,540	\$218,100	\$233,340	0.621	\$144,885
6	—	\$1,500	\$1,800	\$15,540	\$218,100	\$233,340	0.564	\$131,713
7	—	\$1,500	\$2,400	\$15,540	\$218,100	\$232,740	0.513	\$119,433
8	—	\$1,500	\$2,400	\$15,540	\$218,100	\$232,740	0.467	\$108,576
NPV								\$1,186,299

The second dryer is evaluated incrementally by comparing Simulation 3 against Simulation 2.

NPV = \$1,186,299 > \$0 → Future savings exceed the \$60K investment cost. RECOMMEND purchasing the second additional dryer.